

Steve Gwade

Python Backend Engineer

Yaounde, Cameroon · Open to Relocation

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Professional Summary

Python Backend Engineer with 3+ years of cumulated experience designing, building, and deploying scalable distributed software systems using Python, FastAPI, Django, and cloud-native architectures. Specialized in asynchronous processing, event-driven microservices, high-throughput database optimization, and automated continuous integration pipelines. Proven track record of collaborating across cross-functional engineering pods to translate complex business demands and intelligent machine learning models into reliable production backend ecosystems backed by comprehensive testing and observability. Passionate about writing clean, optimized, and maintainable code that solves industrial problems at scale.

TECHNICAL SKILLS

- **Languages:** Python (Asyncio), SQL (PostgreSQL, SQLite), JavaScript, TypeScript, Core Bash Scripting.
- **Backend Architecture:** FastAPI, Django, Flask, REST APIs, Microservices, WebSockets, Distributed Systems, Asynchronous Event Processing.
- **Frontend Integration:** React, TypeScript, HTML5, CSS3, Tailwind CSS framework components.
- **Databases & Caching:** PostgreSQL relational systems, MongoDB, Redis Distributed Caching layers, Data Modeling, Query Optimization.
- **DevOps, Testing & Observability:** Unit Testing, Integration Testing, Observability (Logging, Metrics, Tracing), Docker, Docker Compose, GitHub Actions CI/CD, AWS (EventBridge, SQS queues, EC2, S3).
- **AI & Search Infrastructure:** Large Language Models, Advanced RAG, LangChain, Vector Search, FAISS, PyTorch deep learning systems, TensorFlow.
- **Methodologies & Soft Skills:** Agile/Scrum Development, Technical Stakeholder Alignment, Code Reviews, Cross-Functional Pod Leadership, Algorithmic Analysis, Competitive Programming.

WORK EXPERIENCE

Backend Engineer

Apr 2026 – Present

MITS SARL – Moabi B2B Travel Agency

Yaounde, Cameroon

- Architected highly concurrent asynchronous FastAPI backend webhooks leveraging Python's Asyncio to handle streaming Meta API payloads, slashing message processing latency by 90% ($P_{99} < 120$ ms).
- Designed distributed Redis and DynamoDB caching architectures to securely isolate and manage multi-turn dialogue context states across 500+ concurrent user streams.
- Replaced blocking code paths with decoupled, event-driven backend microservices using AWS EventBridge and AWS SQS queues, allowing the architecture to seamlessly absorb traffic spikes up to 4x baseline capacity.
- Headed a development pod of 2 engineers to deploy automated GitHub Actions CI/CD pipelines with parallel build matrices, reducing deployment lifecycles by 40% while enforcing strict 95%+ Pytest thresholds.

Research Engineer (AI/ML) (Part-time)

Jan 2025 – Jan 2026

JIANTS – Systems Intelligence Lab

Douala, Cameroon

- Profile-tuned 4 multi-modal foundational network architectures using PyTorch Profiler, reducing prototype-to-production deployment latency by 30% across target hardware environments.
- Engineered reproducible PyTorch data processing pipelines featuring automated training telemetry and data versioning via systematic MLflow run tracking configurations.
- Quantized spatial vision layers using TensorRT INT8 post-training protocols, boosting local execution core edge throughput by 56% on parallel CUDA platforms.
- Conducted compute-efficiency profiling (FLOPs, VRAM) and presented empirical benchmarks to technical stakeholders, alignment-guiding upcoming cross-functional execution roadmaps.

Software Engineer (Part-time)

Mar 2024 – Dec 2024

TGNix – IT Consulting

Douala, Cameroon

- Productionalized predictive decision-support scoring models within 2 enterprise ERP modules, implementing automated model monitoring to cut review overhead by 30%.
- Accelerated complex automated decision engines by 25% via targeted SQL query optimization and database performance stress testing across 10M+ transaction records.
- Built high-throughput FastAPI REST endpoints for distributed systems, isolating intensive asynchronous processing workloads across 3 domain-driven microservices.
- Partnered with core engineering pods to containerize multi-tiered application layers via Docker Compose, eliminating environment execution drift across staging platforms.

AI Backend Engineer (Internship)

Jun 2023 – Nov 2023

AITECAF – AI Technologies for Africa

Yaounde, Cameroon

- Engineered Python REST APIs serving 5 core predictive components, negotiating system interface contracts and versioning rules across cross-functional development teams.
- Built modular streaming data pipelines using LangChain, automating structural text parsing and semantic search to

process 50k+ tokens hourly.

- Fine-tuned crop health object detection architectures, optimizing custom anchor box allocations to secure ultra-low sub-5ms edge inference speeds.
- Led data curation cycles and audited annotation metrics for 10k+ high-resolution images, using strict validation splitting rules to eliminate evaluation leakage.

SELECTED PROJECTS

Kite – Cross-Device Scratchpad

Open Source, 2025

- Built a privacy-first, accountless cross-device scratchpad backend using Python and WebSockets to enable secure, real-time data synchronization.
- Designed resilient offline-first data synchronization and state reconciliation mechanisms to maintain document parity and eliminate client-side state conflicts.
- Exposed lightweight, low-latency state propagation REST APIs via FastAPI, ensuring rapid data dissemination across active user sessions.

VERO – Research Workspace for Grounded AI

Open-Source Project, 2025

- Engineered an advanced RAG environment featuring deterministic text parsing, HNSW semantic chunking, and cross-reference metadata generation across document source trees.
- Integrated dense vector retrieval with BM25 matching, optimized via Reciprocal Rank Fusion and Cross-Encoder reranking to maximize retrieval precision.

LINA – Assistive Vision System for the Blind

Huawei ICT Competition 2025

- Co-developed an international multi-modal assistive ecosystem combining streaming object detection and spatial scene comprehension into low-latency execution pipelines.
- Mitigated presentation-attack vulnerabilities by training binary face anti-spoofing classifiers, coordinating with testers to verify performance across variable environments.
- **Awarded 1st Place Worldwide (Innovative Track)** among international entrants for exceptional model compression, serving efficiency, and collaborative architectural novelty.

PlumVision – Real-Time Plum Quality Control Engine

JCIA Hackathon Winner 2025

- Developed an ultra-lightweight 3MB ConvNeXt image classifier using representation learning, registering a 98% accuracy profile on target evaluation datasets.
- Exposed model parameters via optimized FastAPI services, maintaining inference latency under 12 ms on parallel CUDA cores to accelerate system integration.

Multi-Task Transfer Learning for Malaria Diagnosis

MSc Research Thesis, 2025

- Formulated a joint multi-task deep learning problem in PyTorch executing simultaneous semantic cell segmentation and parasite detection from blood smears.
- Evaluated negative transfer optimization dynamics across distinct model heads, establishing a stable joint topology yielding a 97% macro F_1 -score.
- Published a reproducible open-source research repository featuring pipeline configuration matrices, execution scripts, and verified pre-trained model weights for public access.

EDUCATION

MSc in Engineering (Data Science & AI)

2023 – 2025

National Higher Polytechnic School of Douala

Douala, Cameroon

- Graduation Status: Graduated with Distinction (Top 5% of cohort, **GPA: 3.6/4**).
- Core: Mathematical Optimization, Probabilistic Machine Learning, Distributed Deep Learning, Statistical Evaluation.
- **MSc Thesis Repository Link:** github.com/GwadeSteve/MTTL-Research-Malaria

BSc in Engineering (Computer Science)

2020 – 2023

National Higher Polytechnic School of Douala

Douala, Cameroon

- Core Modules: Problem Solving, Algorithmic Complexity, Advanced Data Structures, Linear Algebra, Mathematical Statistics, Systems Engineering.

AWARDS & RECOGNITION

- **1st Place Worldwide Winner**, Huawei ICT Competition 2025 (Innovative Track) – Core Developer for the LINA Assistive Vision project.
- **Ranked 2nd Across Africa**, Orange Digital Center Champions League 2025 – Algorithmic and Competitive Programming tournament (CodinGame).
- **Winner, Best AI Solution** for Optimization and Efficiency, JCIA 2025 Track – Awarded for PlumVision real-time throughput and deployment speed.
- **Winner, Most Innovative AI Prototype**, JCIA 2025 Track – Awarded for EatWise, a multi-modal computer vision framework for targeted nutritional parsing.